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ROBOTICS ENGINEER

Manufacturing Process Plan Simulation – NASA Bracket 478019

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1. INTRODUCTION

1.1. PROJECT BACKGROUND AND OBJECTIVES

This Manufacturing Process Plan (MPP) aims to investigate and develop a technically feasible manufacturing process for an aerospace-related component through reverse engineering.

This study is conducted exclusively for educational purposes, with the objective of demonstrating the application of manufacturing engineering principles, Computer-Aided Design (CAD), Computer-Aided Manufacturing (CAM), and CNC programming using Siemens NX.

The project focuses on developing and justifying a modern CNC manufacturing strategy based on the component's original engineering documentation.

The main objectives of the MPP are:

1. **Component Selection**: Identify and select a suitable mechanical component originally developed for an aerospace or related application
2. **Component Engineering Analysis**: Examine the component's dimensions, geometry, tolerances, functional features and manufacturing requirements.
3. **Applicable Standards**: Identify and interpret relevant engineering and manufacturing standards, while preserving the specifications and tolerancing conventions of the original technical drawing.
4. **Material Selection**
5. **Manufacturing Process Planning**: Establish a technically justified manufacturing strategy.
6. **CNC Machine and Workholding Selection**
7. **Raw Stock Selection:** Determine the required stock geometry and dimensions, accounting for machining allowances, material tolerances and workholding requirements.
8. **Cutting Tool and Machining Parameter Selection:** Select Suitable cutting tools and calculate the required cutting speeds, spindle speeds, feed rates and depths of cut.
9. **CAD/CAM Programming:** Reconstruct the component geometry and develop the required CNC machining operations using Siemens NX CAD/CAM
10. **Machining Verification and Simulation:** Verify the proposed machining operations through toolpath simulation, material-removal analysis, and collision checking.
11. **Dimensional Inspection and Quality Control:** Establish inspection procedures, measurement methods, and acceptance criteria to evaluate conformity with the engineering drawing.
12. **Finishing and Post-Processing:** Define the necessary deburring, surface finishing, cleaning and protective treatment procedures according to the component's requirements.

The expected outcome is a documented and technically justified manufacturing process that demonstrates the systematic application of engineering principles, from the interpretation of an original aerospace engineering drawing to the development and virtual verification of a CNC machining process.

1.2 SCOPE AND LIMITATIONS

The scope of this document is to develop and document a comprehensive manufacturing process plan for an aerospace or related mechanical component using Siemens NX for CAD/CAM

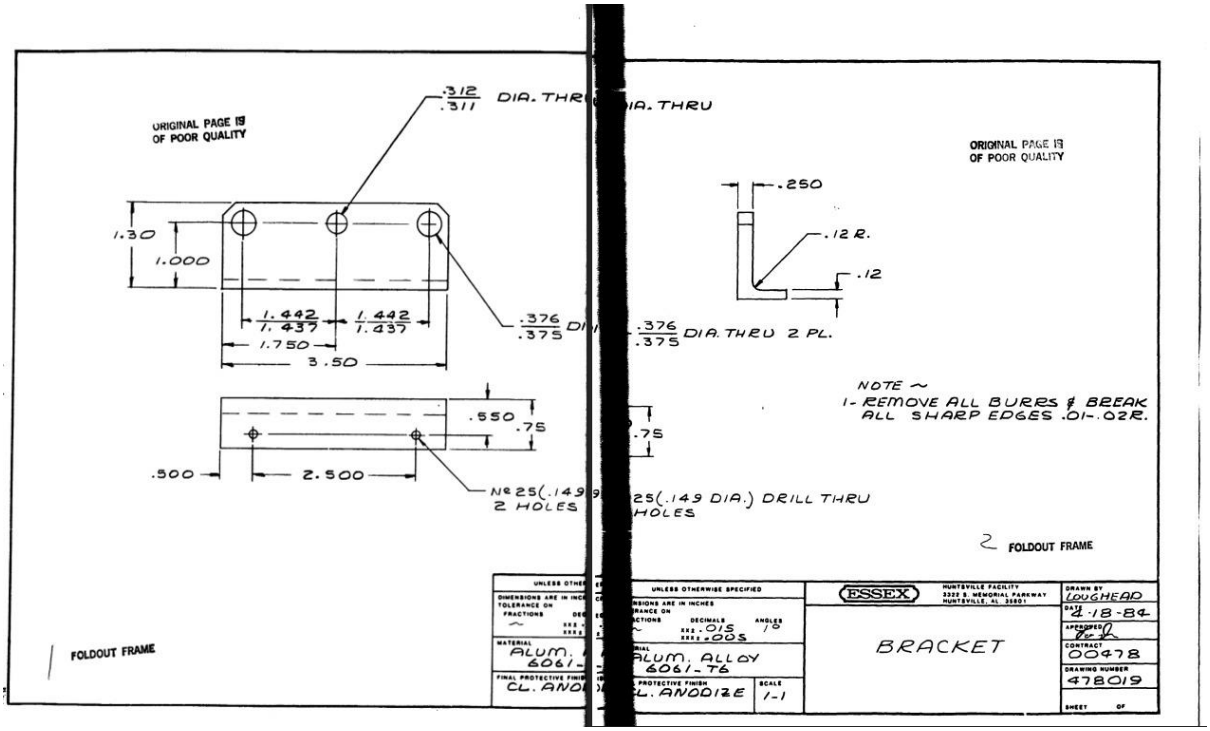
The component will be selected from the engineering drawings included in the technical report Development of Robotics Facility, Docking Test Hardware, prepared under NASA Contract NAS8-34656. The report, publicly available through the NASA Technical Reports Server (NTRS), documents the development and fabrication of hardware for a robotic docking test facility.

The project aims to reconstruct a selected component from its original engineering drawing and develop a feasible manufacturing process using a modern CNC machining center. The original dimensions, tolerances and applicable technical requirements will be preserved, including the inch-based dimensioning and tolerancing system. Geometric features and manufacturing requirements not explicitly defined in the original drawing will be established through justified engineering assumptions and identified separately from the original specifications.

To ensure sufficient complexity for demonstrating CAD/CAM and CNC programming skills, the following criteria were established for selecting a suitable component:

- At least one fillet or rounded transition
- At least one chamfered edge
- At least two holes with different nominal diameters
- Geometry features requiring multiple machining operations or orientations

After an examination of the available drawings, **Bracket 478019 was selected as the subject of this study.**



The manufacturing process plan will cover the analysis of the component geometry, material selection, machining process planning, CNC machine selection, workholding, raw stock determination, cutting tool selection, machining parameter calculation, and the development and verification of machining operations using Siemens NX CAM. Dimensional inspection methods and quality-control requirements will also be considered.

Project limitations: The objective is to develop a technically justified manufacturing process based on the original component specification, rather than reproduce the exact historical manufacturing procedures used by the original contractor. Machining parameters and simulated results will require physical validation before the

process can be considered production-ready. The resulting component is intended for engineering demonstration purposes and is not represented as qualified aerospace flight hardware.

1.3 APPLICABLE TECHNICAL STANDARDS

The development of this Manufacturing Process Plan (MPP) will be guided by internationally recognized engineering, manufacturing, and quality assurance standards. These standards provide a technical framework for interpreting engineering drawings, selecting materials, planning machining operating, programming CNC equipment, and establishing inspection procedures.

As the selected component, Bracket 478019 originates from an engineering drawing produced by Essex Corporation under NASA Contract NAS8-34656 in 1984, particular attention will be given to preserving the original design specifications.

The original engineering drawing will serve as the primary reference for all dimensional, geometric, material, and finishing requirements. Modern standards will be used to support the proposed methodology without modifying the original specifications.

1.3.1 ORIGINAL ENGINEERING SPECIFICATION

The primary technical reference for this project is **Drawing 478019** included in the mentioned document.

The drawing establishes the following manufacturing requirements:

- **Dimensional system:** Imperial units (inches)
- **Material:** Aluminum alloy 6061-T6
- **General dimensional tolerances:**
    - Two-decimal dimensions: ±0.015 in
    - Three-decimal dimension: ±0.005in
- **General angular tolerance:** $\pm 1^{o}$
- **Specific dimensional tolerances:** Explicitly defined limits for certain hole diameters.
- **Edge treatment:** Removal of burrs and breaking of sharp edges to a radius of 0.01-0.02 in.
- **Protective surface finish:** Clear anodizing.

Individual dimensions with explicitly specified tolerance limits will take precedence over general drawing tolerances.

The original specifications will be maintained throughout CAD modelling, manufacturing process development, and dimensional inspection.

1.3.2 ENGINEERING DRAWING AND DIMENSIONAL STANDARDS.

The following standards will be used as supplementary references for engineering drawing interpretation, CAD documentation, and dimensional verification

Standard	Description	Application
ASME Y 14.5-2018 (R2024)	Dimensioning and Tolerancing	Reference for dimensional and geometrical tolerancing principles
ASME Y 14.100-2017	Engineering Drawing Practices	Preparation and revision control of engineering drawings
ASME B46.1-2019	Surface Texture	Reference for surface roughness characterization when applicable

These standards will guide the preparation of newly developed technical documentation and will not be applied retroactively where their requirements could alter the interpretation of the original 1984 drawing.

1.3.3 MATERIAL STANDARDS

Material selection and procurement will be supported by ASTM specification applicable to the chosen aluminum stock form

Standard	Description	Application
ASTM B209/B209M	Aluminum and Aluminum Alloy Sheet and Plate	Material specification if rolled plate stock is selected
ASTM B221	Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes	Material specification if extruded bar stock is selected

The applicable material standard will be determined after establishing the manufacturing strategy, selecting the workholding method, and calculating the required stock dimensions.

Material documentation and supplier specifications will be examined to confirm the selected alloy, temper, dimensions, and relevant mechanical properties.

1.3.4 CNC MANUFACTURING AND MACHINE SAFETY STANDARDS

The following international standards will support CNC programming and manufacturing safety considerations.

Standard	Description	Application
ISO 6983-1:2009	Numerical Control of Machines: Program Format and Definitions of Address Words	Reference for CNC program structure and numerical control programming
ISO 16090-1:2022	Machine Tools Safety: Machining Centers, Milling Machines, Transfer Machines	Reference for machining center safety requirements and protective measures

CNC machining operations will be developed using Siemens NX CAM, and the resulting toolpaths are recommended to be postprocessed using a compatible Haas postprocessor to generate NC programs for the Haas Next Generation Control (NGC) system of the selected VF-2SS machining center.

The generated NC programs will be verified against the machine's actual control configuration, programming documentation, travel limits and available functions.

1.3.5 QUALITY ASSURANCE AND INSPECTION STANDARDS

Although the project is educational and does not involve the production of certified hardware, recognized aerospace quality standards will be used as references for the development of inspection and documentation practices.

Standard	Description	Application

ISO 9001:2026	Quality Management Systems: Requirements	Reference for documentation, traceability, and process control
AS9100D	Quality Management Systems: Requirements for Aviation, Space, and Defense Organizations	Reference for aerospace manufacturing quality practices
AS9102C:2023	Aerospace First Article Inspection Requirements	Reference for documenting dimensional inspection and component conformity

Particular emphasis will be placed on AS9102C, which provides a structured approach to verifying and documenting the characteristics of a manufactured component against its engineering requirements.

An inspection plan will be developed following the principles of AS9102C, defining the required inspection characteristics, measurement methods, and procedures for verifying dimensional and surface quality. A simplified First Article Inspection (FAI) report structure will be proposed, without actual measurement results, as the manufacturing process is evaluated virtually.

1.3.6 SURFACE TREATMENT STANDARDS

The original drawing specifies a clear anodized protective finish.

For the proposed manufacturing process, MIL-PRF-8625F, Anodic Coatings for Aluminum and Aluminum Alloy will be considered as a reference for developing a suitable anodizing specification. Given that the original drawing does not specify the anodizing procedure, choosing MIL-PRF-8625F is a modern process selection

The appropriate anodizing parameters will be determined during the finishing and post-processing stage.

1.3.7 APPLICATION AND LIMITATIONS OF STANDARDS

The standards identified in this section serve as a technical reference. In the event of conflicting requirements, the original drawing and its specified tolerances with take precedence for the purposes of the educational reconstruction.

Standards introduced to support modern manufacturing, quality assurance, or documentation will be applied only where relevant and technically justified.

The use of these standards does not constitute formal certification, qualification, or evidence of compliance with NASA aerospace manufacturing requirements. The project is intended for demonstration purposes regarding engineering principles and a virtually verified CNC manufacturing process.

2. COMPONENT ENGINEERING ANALYSIS

2.1 COMPONENT OVERVIEW

The selected component, Bracket 478019 is an L-shaped aluminum mounting bracket documented in Appendix C of the document presented in section 1.2. The bracket was developed as part of the experimental docking hardware designed for NASA's Marshall Space Flight Center. Its precise function is not explicitly described in the report, but its geometry suggests that is serves as a structural mounting or connection element between mechanical components.

The objective of this analysis is to identify the components' geometric characteristics, dimensional requirements, and potential manufacturing challenges as a basis for developing a suitable CNC manufacturing process.

2.2 GEOMETRIC AND TECHNICAL SPECIFICATIONS

The component has an L-shape consisting of two perpendicular walls connected with an internal fillet. Its geometry includes five through-holes of three different nominal diameters and chamfered external corners.

After carefully analysis the original image, the following characteristics have been identified:

Feature	Specification
Overall dimensions	3.50 x 1.30 x 0.75 in
Bottom wall (No.25 holes)	0.120 in (3.048 mm)
Main wall (3 mounting holes)	0.250 in (6.35 mm)
Internal fillet	R 0.120 in (3.048 mm)
Chamfer	Not specified
Larger mounting holes	2 x Ø0.375-0.376 in (9.525 – 9.55 mm)
Central mounting hole	1 x Ø0.311-0.312 in (7.899 – 7.925 mm)
Smaller mounting holes	2 x No.25 drill (Ø0.1495in / 3.7973 mm nominal)

The original drawing also establishes general dimensional tolerances:

- Two-decimal dimensions: ±0.015 in
- Three-decimal dimensions: ±0.005 in
- Angular dimensions: ±1°
- Individual dimensions with explicitly specified limits take precedence over general tolerances.

After experimental procedures in the CAD software, a chamfer of 0.15 in x 45° was chosen as it visually matches the drawing and corresponds to other technical characteristics.

No numerical surface roughness requirement or explicit modern GD&T specifications were identified in the original drawing. For the purposes of this project, a Roughness of Ra $\leq$ 3.2 µm (125 µin) was chosen for general machined surfaces, and Ra $\leq$ 1.6 µm (63 µin) for flat mounting surfaces and precision holes. Keep in mind that these roughness values are not universal aerospace standards, but a practical choice for this project.

The drawing requires the removal of burrs, breaking sharp edges to a radius of 0.01-0.02 inches and applying a clear anodized protective finish.

Original material is aluminum alloy 6061-T6. Its properties machinability, and commercial availability will be examined in Chapter 3.

2.3 MANUFACTURING CONSIDERATIONS

The geometry of Bracket 478019 introduces several manufacturing challenges. The thin walls, particularly the 0.120 inch section, require careful machining and workholding to minimize deformation. The internal R 0.120 inch fillet must be considered when selecting cutting tools, while the holes located on perpendicular faces require different machining orientations.

Additionally, the closely tolerance holes (0.001-inch diameter range) demand accurate machining and inspection. These factors will influence the selection of machining operations, cutting tools, and workholding arrangements in subsequent chapters.

3. MATERIAL SELECTION

3.1 ORIGINAL MATERIAL SPECIFICATION

The original engineering drawing of Bracket 478019 specifies Aluminum Alloy 6061-T6, a precipitation-hardened aluminum alloy primarily containing magnesium and silicon. The T6 designation indicates that the material has been solution heat-treated and artificially aged to improve its mechanical strength.

6061-T6 is commonly used in aerospace, marine, and general mechanical engineering applications. Typical components include brackets, mounting plates, machine frames, and support elements.

For the selected bracket, the alloy offers a good combination of low weight and mechanical strength, combined with corrosion resistance and machinability, making it suitable for a structural component within experimental docking hardware.

3.2 MECHANICAL AND PHYSICAL PROPERTIES

The principal material properties of 6061-T6 aluminum are presented below.

Property	Typical value
Density	2.70 g/cm^3
Ultimate tensile strength	310 MPa
Yield strength (0.2%)	276 MPa
Young's modulus	68.9 GPa
Elongation at break	12-17 %
Brinell hardness	95 HB
Poisson's ratio	0.33
Thermal conductivity	167 W/(m·K)
Melting range	582-652 ^{o}C

Source: MatWeb/ASM

3.3 MATERIAL MACHINABILITY

6061-T6 aluminum exhibits good machinability and is well suited for CNC milling and drilling. Its relatively low cutting resistance permits efficient material removal, while its mechanical properties allow accurate machining with suitable tooling and workholding.

Sharp, polished carbide cutting tools are generally recommended for aluminum machining. Also, good chip evacuation and appropriate coolant or lubrication will be selected to prevent built-up edge and maintain consistent surface quality.

The alloy responds well to anodizing, making it compatible with the original end treatment mentioned.

3.4 FINAL SELECTION AND PROCUREMENT

6061-T6 is a widely available commercial alloy supplied in various forms. For this project, solid rectangular stock will be considered. The final stock dimensions will be determined after establishing the machining strategy and workholding requirements. **As the original material remains commercially available, no material substitution is necessary.**

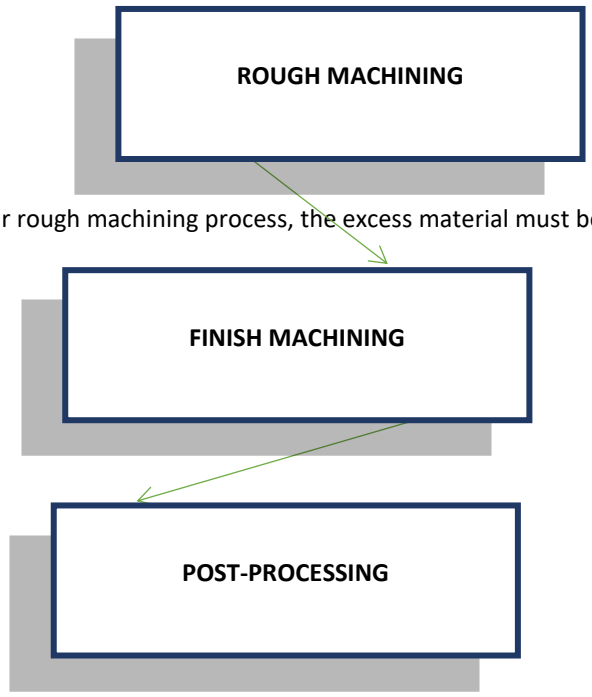
4. MANUFACTURING PROCESSES AND STRATEGY

4.1 MANUFACTURING STRATEGY

The manufacturing process for Bracket 478019 will be developed using subtractive CNC machining, with the component manufactured from a solid aluminum 6061-T6 block.

For the manufacturing of the part, a CNC with at least 3 axis is needed, and given the L-shape figure of the part, the mounting position will be need to change as well. Also, the machine should have a magazine with sufficient tool space to support all operations. Given the small dimensions of our bracket, the available working size will not be an impediment. **The selected machine is Haas VF-2SS, as explained in Chapter 5. Siemens NX will be used to develop the machining simulation.**

The proposed strategy consists of 3 main stages:



At every step of our rough machining process, the excess material must be sufficient for finishing and workholding. The Rough machining phase will remove the majority of the material to establish the L-shaped geometry. The finish machining process will reduce the material to the required dimensions and produce the internal radius, mounting surfaces and holes within tolerances and roughness limits. During post-processing, burrs will be removed and the specified protective finish will be applied.

4.2 REQUIRED MANUFACTURING PROCESSES

According to the geometric features and technical requirements, the following processes have been established.

Manufacturing process	Application	Proposed Siemens NX CAM operation
1. Face milling	Establishing flat reference and mounting surfaces	Face milling
2. Rough milling	Removing material to create the L-shaped cross-section. Two operations: internal L-profile and external perimeter	Cavity Mill / Adaptive Milling
3. Wall finishing	Achieving final wall thickness and vertical surface quality	Floor Wall / Planar Mill
4. Profile milling	Finishing external contours and component's overall dimensions	Profile Milling
5. Radius finishing	Producing the R0.120-inch internal fillet	CONTOUR_AREA or suitable form-tool strategy
6. Spot drilling	Preparing accurately located holes	Spot Drilling
7. Drilling	Producing the initial through holes	Drilling
8. Precision hole finishing	Achieving the specified diameter limits of the three larger holes	Hole Milling / Boring / Reaming
9. Edge finishing	Removing burrs and producing specified edge radii	Controlled deburring / edge rounding

The geometry can primarily be manufactured using 2.5-axis milling operations.

The three larger holes introduce an additional level of complexity as they have particularly restrictive diameter limits, with a tolerance of just 0.001 inches. Consequently, drilling alone will not be considered sufficient to guarantee the required dimensions and a precision finishing operation will be selected after evaluating tool availability.

The two smaller holes are specified as No. 25 drilled through-holes and may be produced using an appropriate standard drill.

4.3 PROPOSED MACHINING SEQUENCE

The manufacturing process will be organized into three provisional CNC setups. The final setup sequence may be adjusted depending on the selected fixture and raw stock geometry.

Setup 1: Primary Machining (G54)

The stock will be positioned to allow machining of the main L-shaped cross section and access to the three larger mounting holes.

The proposed operations are:

1. Face the upper surface to establish a machining reference
2. Rough mill the material into L-shaped profile
3. Perform rest machining where material remains inaccessible to the primary roughing cutter
4. Finish the horizontal and vertical surfaces

5. Finish the R 0.120 inch internal fillet
6. Spot drill and drill the three larger mounting holes
7. Finish the holes using a suitable precision machining strategy.

Sufficient material must be retained during roughing to support the thin sections and permit subsequent finishing operations.

Setup 2: Reverse-Side Machining (G55)

In this setup, the excessive material that remained during previous operations will be removed, including material used for workholding or unfinished stock material.

Depending on the stock and previous operations, the setup might include:

1. Removing sacrificial material retained for workholding
2. Face milling the underside to final thickness
3. Finishing the remaining external surfaces
4. Opening or completing through holes where temporary supporting material was retained
5. Removing remaining machining burrs.

Setup 3: Perpendicular Drilling (G56)

The component will be repositioned by 90° to allow the drilling of the two smaller holes on the bottom.

The proposed operations are:

1. Establish the secondary machining coordinate system.
2. Verify the part position and clamping.
3. Spot drill the two holes.
4. Drill the two No.25 through-holes.
5. Deburr the accessible hole entrances.

The workholding arrangement must provide adequate support without deforming the thin walls.

The necessity and sequence of this setup will be reassessed during workholding and stock selection.

Preliminary Manufacturing Routing

Operation	Process	Objective
OP010	Material Inspection	Verify material and stock condition
OP020	Saw cutting	Prepare raw billet
OP030	Face milling	Establish reference surface
OP040	Rough milling	Create L-shaped profile
OP050	Rest milling	Remove remaining excess material
OP060	Finish milling	Finish walls and planar surfaces
OP070	Radius finishing	Produce internal R0.120 inch fillet
OP080	Spot drilling and drilling	Prepare three larger holes
OP090	Precision hole finishing	Achieve specified hole diameter limits

OP100	Repositioning	Establish second machining orientation
OP110	Reverse-side machining	Complete underside and remaining features
OP120	Repositioning	Establish third machining orientation
OP130	Drilling	Produce two No.25 holes
OP140	Deburring and edge rounding	Remove burs and treat sharp edges
OP150	Dimensional and roughness inspection	Verify the required characteristics
OP160	Clear Anodizing	Apply protective surface
OP170	Final inspection	Verify the finished component

This sequence is preliminary and will be refined during detailed CNC programming.

4.4 SURFACE FINISH STRATEGY

The machining process must satisfy both the original drawing requirements and additional roughness specifications added specifically for this project.

The following roughness values will be applied:

SURFACE CATEGORY	**MAXIMUM ROUGHNESS**
GENERAL MACHINED SURFACES	Ra 3.2 µm (125 µin)
DESIGNATED FUNCTIONAL MOUNTING PLATES	Ra 1.6 µm (63 µin)
PRECISION HOLE SURFACES	Ra 1.6 µm (63 µin)

These roughness requirements are based on the surface texture principles described in ASME B46.1 and will apply before anodizing.

Rough machining operations will prioritize leaving sufficient stock for finishing operations. Dedicated finishing passes will be used to obtain the required dimensions and surface quality.

The original drawing requires the removing of all burs and breaking sharp edges to a radius of 0.01-0.02 inches. This requirement will be treated separately and where necessary, controlled manual deburring or mechanical edge-rounding operations will be used.

4.5 MANUFACTURING CONSTRAINTS AND PROCESS VERIFICATION

The following factors will be considered when developing and validating the manufacturing process:

Thin-wall rigidity: The thin 0.120 inch section might be susceptible to deformation caused by clamping forces or excessive cutting loads. The machining sequence and workholding setup must provide sufficient support to prevent the deformation during the removal of material

Geometric accessibility: The perpendicular shape of the mounting faces requires the repositioning of the component. Machining setup must provide adequate tool and holder clearance.

Dimensional accuracy: The fine tolerances of the holes require controlled finishing processes and suitable inspection methods.

Surface quality: Roughing and finishing operations must be planned separately where necessary for achieving the required roughness.

Manufacturing verification: All machining operations will be programmed and verified in Siemens NX. The generated NC program will require verification against the selected Haas machine configuration.

The final machining sequence, tools, parameters etc. will be established in subsequent chapters.

5. CNC MACHINE SELECTION

5.1 SELECTED CNC MACHINE

The selected machine for manufacturing Bracke 478019 is the Haas VF-2SS, because it is a compact vertical CNC machine center designed for high-speed milling, drilling, tapping and precision machining of small to medium components.

The machine is equipped with a 31-position tool changer.

Here are the principal technical specifications, listed on the official as of October, 2026:

PARAMETER	SPECIFICATION
MANUFACTURER	Haas Automation, Inc.
MODEL	VF-2SS (Super-Speed)
MACHINE CONFIGURATION	3-axis vertical machining center
CNC CONTROLLER	Haas Next Generation Control (NGC)
X/Y/Z TRAVEL	762 x 406 x 508 mm
WORKING TABLE	914 x 356 mm
MAXIMUM TABLE LOAD	680 kg
SPINDLE SPEED	12,000 rpm;
MAXIMUM SPINDLE POWER	22.4 kW (30HP)
MAXIMUM SPINDLE TORQUE	122 Nm at 2,000 rpm
TOOL MAGAZINE CAPACITY	30 + 1
TOOLHOLDER INTERFACE	CT40; BT40 or HSK-A63 configurations available
MAXIMUM CUTTING FEED RATE	21,200 mm/min
RAPID TRAVERSE	35.6 m/min (X, Y, Z)
POSITIONING ACCURACY	Not specified
REPEATABILITY	Not specified
MACHINE WEIGHT	Approx. 3,539 kg (listed shipping weight)

5.2 MACHINE SUITABILITY

The Haas VF-2SS was selected primarily for its compact dimensions, machining versatility and suitability for aluminum components.

Bracket 478019 has approximately 88.9 x 33.0 x 19.1 mm, comfortably suiting the machine's envelope. The spindle speed of the CNC combined with its carbide tools are suitable for machining 6061-T6 aluminum and supports most required operations.

5.3 MACHINE COST AND AVAILABILITY

Haas VF-2SS machining centers are widely available commercially as both new and used equipment. As a According to the manufacturer's published pricing, a new Haas VF-2SS has a starting price of **73.995 €, depending on the selected configuration and optional equipment.**

This figure represents and indicative manufacturer-listed price, rather than a formal quotation. Actual acquisition costs may vary by region and do not include things such as transportation, installation, taxes, tooling, etc. Used machine's prices vary depending on multiple factors.

This project will treat this equipment as a representative industrial CNC machining center rather than an equipment purchase requirement.

6. MACHINING SETUP AND WORKHOLDING PLANNING

6.1 WORKHOLDING STRATEGY

The manufacturing process will involve three different machining setups using different component orientation to ensure accessibility and rigidity during machining.

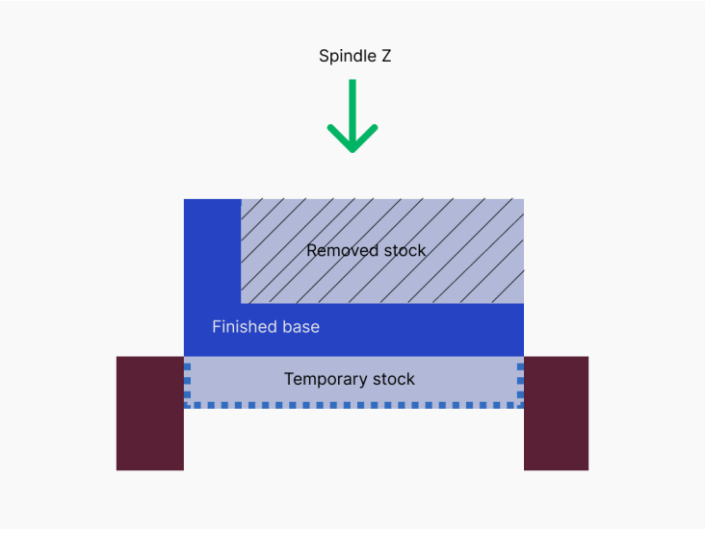
A precision machine vise with custom jaws and supporting fixtures will be used to secure the component, as its thin walls are susceptible to be affected by deformation caused by cutting and clamping forces.

6.2 MACHINING SETUPS

Setup 1: Primary Machining

The initial billet will be clamped in a vise using sacrificial material beneath the L-shaped form. This arrangement provides rigidity and allows for the drilling of the three main holes and the removal of most excess material.

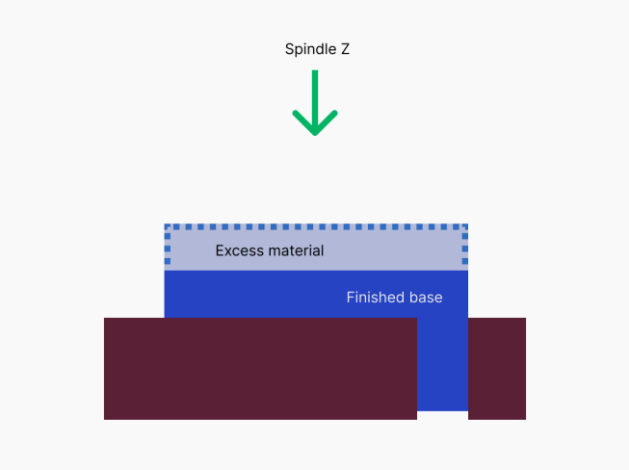
Main operations include rough milling the L-shaped profile, finishing the walls and interior radius and drilling the mounting holes.



Setup 2: Reverse-Side Machining

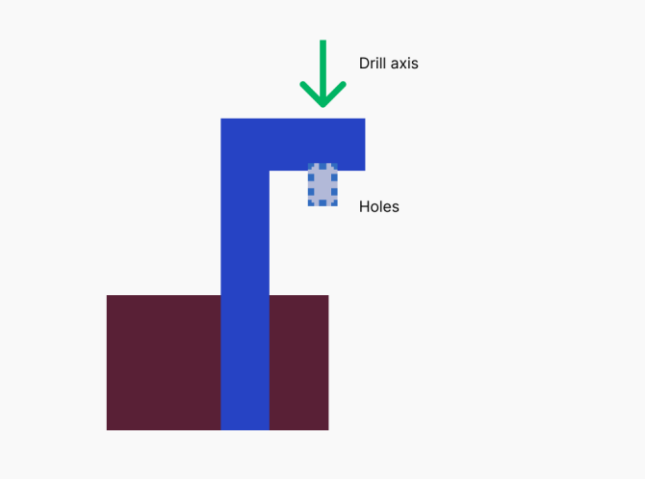
The component will be inverted 180° and secured using custom jaws and a supporting fixture for the removal of the surplus material. The think long wall will be supported along its length to prevent deflection during machining.

The primary operation consists of removing the sacrificial material and finishing the underside to its final thickness. The fixture must provide sufficient resistance to both vertical and lateral cutting forces.



Setup 3: Perpendicular Hole Machining

The component will be repositioned to allow the drilling of the No. 25 drill, nominal Ø0.1495 inch (3.7973 mm). A custom fixture will support and secure the component from both sides, preventing the movement during drilling. The fixture will also provide clearance for drill breakthrough and avoid excessive clamping forces on the thin sections.



6.3 WORKHOLDING AND SETUP VERIFICATION

The proposed workholding arrangements establish the preliminary clamping and support requirements for each machining setup.

These requirements will be considered during raw stock selection, for example, when determining the sacrificial material for Setup 1.

After stock selection, the fixture dimensions and clamping arrangements will be finalized and verified in Siemens NX to assess tool accessibility and potential collisions. Adequate workholding rigidity will require separate mechanical verification before physical manufacturing.

7. STOCK SELECTION AND ALLOWANCE CALCULATIONS

7.1 STOCK SELECTION METHODOLOGY

The raw stock dimensions are determined based on the component's finished geometry and the operations and strategies established in Chapters 4 and 6.

A solid rectangular billet of aluminum 6061-T6 is selected to accommodate the machining of the L-shaped surface while providing sufficient material for clamping in the first phase.

Machining allowances, workholding allowances, material tolerances, and material availability should be taken into account during stock selection. ASTM B221 covers extruded aluminum bars and profiles, while ASTM B209/B209M applies to aluminum plate. The applicable material specification will depend on the stock form supplied.

Neither specification establishes a universal machining allowance for the finished bracket. These values must be determined according to the manufacturing process, stock condition and fixture requirements.

7.2 MACHINING ALLOWANCES

Machining allowances are selected to provide sufficient material for producing the final dimensions and surface quality. The following values are preliminary in calculating the required 6061-T6 aluminum stock:

Operation	Proposed Allowance	Justification
Initial face milling	1.0 mm	Remove the original stock surface and establish a reference
External side and end finishing	1.0 mm per side	Allow squaring and dimensional finishing
Rough milling of L-profile	0.5 mm on walls and 0.3 mm floors	Leave material for finish milling
Final wall/floor milling	Remove remaining 0.5/0.3 mm	Achieve final dimensions and roughness
Precision hole finishing	Tool-dependent	Achieve the specified hole diameter limits
Reverse-side machining	Remaining sacrificial stock	Remove clamping material and finish the underside
Edge rounding	R0.01-0.02in	Meet the original drawing's edge-treatment requirement

The roughing allowances of 0.5 mm on walls and 0.3 mm on floors are included within the material already allocated to the final geometry and **are not added separately to the external stock dimensions.**

7.3 PRELIMINARY CUTTING-FORCE ANALYSIS

In order to determine the approximate mechanical loads acting on the workpiece during rough milling, a preliminary cutting-force analysis is required. Particular attention is given to the initial machining setup (G54), in which the workpiece is secured through a sacrificial stock extension.

The analysis will be based on the cutting geometry and the recommended machining parameters of a selected solid-carbide end mill suitable for the chosen Aluminum 6061-T6. HSMAdvisor will be used to estimate cutting force, spindle torque, cutting power, and tool deflection under the proposed cutting conditions.

The calculated cutting loads will be subsequently evaluated in relation to the toolpath direction, the location of the cutting zone and the workpiece clamping arrangement. The resulting force components and moments will provide a preliminary basis for assessing resistance to sliding and rotation and deformation during machining.

During a preliminary force analysis, these are the resulting parameters obtained using HSMAdvisor. These parameters confirm that a Ø10mm carbide mill is suited for the roughing operation. A suitable mill will be selected from a manufacturer's catalog and its specifications will be used in calculating the estimated preliminary forces.



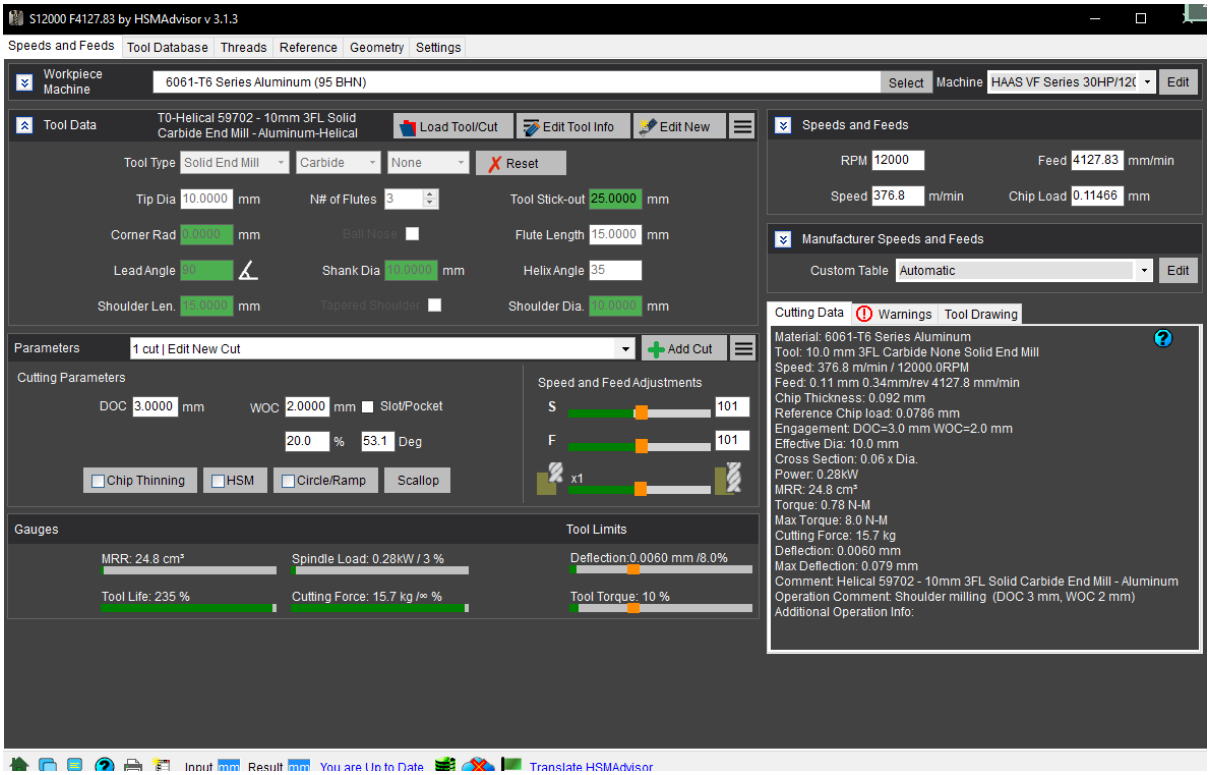
Therefore, a Ø10mm end mill from Helical Solutions, full name: Helical Solutions MH35AL-015-31000 solid-carbide end mill (tool No. 59702) with three cutting edges was selected for the analysis. We kept our Haas machine configuration with 30HP and a maximum 12,000 RPM, an axial depth cut of 3 mm and a radial engagement of 2 mm.

The software calculated a spindle speed of 12,000 rpm, a feed rate of 4,087 mm/min, and a material removal rate of approximately 24.5 cm^3/min. Under these conditions, HSMAdvisor predicted a cutting force of approximately 153N, a cutting torque of 0.77 Nm, and a tool deflection of 0.0059 mm. The reported cutting power was 0.28 kW. Because the reported power and torque do not correspond to the same steady rotational load, their interpretation requires further verification.

A unit conversion confirms that the cutting force reported by HSMAdvisor corresponds to approximately 153N.

$$F = 15.6 \times 9.860665 = 153\,N\ (reported\ force)$$

These values constitute preliminary estimates for assessing the machining process. The predicted force must subsequently be evaluated with respect to its direction, application point and resulting moments acting on the workpiece and clamping arrangement.



To evaluate workholding requirements, the estimated cutting force must first be resolved into components acting relative to the tool and workpiece. In peripheral milling, the cutting load can be represented by tangential (Ft), radial (Fr) and axial (Fa) components. **The tangential components is associated with the primary cutting action, while the radial and axial components depend on cutter geometry, engagement and chip formation.**

Singe HSMAdvisor do not provide individual force components, the following preliminary assumptions were adopted:

$$Ft = 154\ N$$

$$Fr = 0.50 \times\ Ft = 77N$$

$$Fa = 0.30 \times Ft = 46.2\ N$$

These ratios are not experimentally verified, but are used to provide an approximate model for the theoretical workholding analysis.

The resultant force acting in the horizontal machining plane is calculated as:

$$Fh = \sqrt{(Ft^2 + Fr^2)} = \sqrt{(154^2 + 77^2)} = 172.2\ N$$

The actual force direction changes during the milling process. To account for uncertainties, a design load factor of 2.0 is introduced.

$$F_{hd} = 2 \times 172.2 = 344.4\,N \approx 350\,N$$

$$F_{ad} = 2 \times 46.2 = 92.4\,N \approx 100\,N$$

Therefore, 350 N and 100 N are adopted as preliminary and theoretical horizontal and axial design load. Their applicability requires maintaining the assumed 2 mm radial engagement through an appropriate roughing strategy.

Preliminary workholding analysis.

In setup G54, the initial billet is clamped through a sacrificial extension located under the finished base. The configuration allows the main part geometry to be machined without applying the vise pressure directly to the thin finished features. For the theoretical analysis, a vise with machinable soft jaws is considered, with the following assumed parameters:

Parameter	Value
Clamping force per jaw	3,000 N
Effective friction coefficient	0.15
Jaw contact length	90 mm
Jaw engagement height	12 mm

The frictional holding capacity of the two opposing haws is estimated as:

$$F_{friction} = 2 \times \mu \times N = 2 \times 0.15 \times 3000 \approx 900\,N$$

$$F_{shear} = \sqrt{(350^2 + 100^2)} = 364\,N$$

The theoretical friction-capacity ratio is therefore:

$$R_{sliding} = \frac{900}{364} = 2.47$$

This suggests sufficient resistance to sliding under the assumed conditions. However, a positive positioning stop and adequate support are recommended to reduce dependence on friction.

Overturning Moment and Jaw Engagement

Because the cutter operates above the vise jaws, the horizontal cutting force generates and overturning moment. Assuming a 0.5 mm clearance between the finished base and the upper jaw contact, the distance from the highest cutting point to the center of the 12 mm jaw engagement is:

$$h = 19.05 + 0.5 + \frac{12}{2} = 25.55\,mm$$

$$Move_{r_turning} = 350\,x\,0.02555 = 8.92\,Nm$$

Assuming the moment is distributed equally between the two jaw contacts, the resulting load eccentricity is:

$$e = \frac{\frac{8.94}{2}}{3000} = 1.49\,mm$$

For an idealized linear-contact pressure distribution, the condition for maintain compressive contact is:

Since 1.49 mm is below 2.00 mm, the proposed haw engagement is considered reasonable for preliminary sizing, but this simplified assessment does not replace verification of actual jaw contact, stiffness, and fixture geometry.

Sacrificial Stock Height

Based on the assumed workholding arrangement: the sacrificial extension must accommodate:

- 12 mm of effective jaw engagement;
- 0.5 mm clearance beneath the finished base;
- 2.5 mm additional material for positioning and support.

The resulting height is:

$$H_{sacrificial} = 12 + 0.5 + 2.5 = 15.0\,mm$$

The 2.5 mm allowance is a geometric design assumption. Considering the finished component height of 19.05 mm and an additional 1 mm top-facing allowance:

$$H_{stock} = 19.05 + 15.00 + 1.00 = 35.05\,mm$$

For preliminary stock selection, a nominal height of 40 mm is considered, providing approximately 19.95 mm of sacrificial material after the assumed facing operation. Therefore an **approximately 20 mm sacrificial extension is adopted for the theoretical manufacturing strategy.**

7.4 STOCK DIMENSION CALCULATIONS

Based on the preliminary cutting-force and workholding analysis presented in the previous section, the initial stock dimensions can be established. The billet must take into account the finished bracket geometry, machining allowances, and the sacrificial extension required for clamping in the first setup.

The overall dimensions of the finished component, converted to mm are:

Dimension	Finished Size
Length (X)	88.90 mm
Width (Y)	33.02 mm
Height (Z)	19.05 mm

Length and Width Calculations

A nominal billet of 100 mm length x 40 mm width is proposed, leaving a total X allowance of 11.10 or 5.55 mm/side, and a Y allowance of 6.98 or 3.49 mm/side. These allowances provide sufficient material for machining the external surfaces and compensating for initial stock irregularities.

The height of the stock, as mentioned in the previous chapter, will be 35.05 mm. A nominal billet height of 40 mm is proposed. After removing 1 mm during initial facing, the sacrificial height becomes 19.95 mm.

Stock Volume and Mass

The proposed nominal billet dimensions are 100 x 40 x 40 mm, therefore the stock volume will be:

$$V_{stock} = 100 \times 40 \times 40 = 160{,}000\ mm^3 = 160\ cm^3$$

Using the approximate density of aluminum 6061-T6 (2.70 g/cm^3), the mass is obtained:

$$m = \rho_{6061-T6} \times V_{stock} = 2.70 \times 160 = 432\ g\ =\ 0.432\ kg$$

Stock Selection Summary

Parameter	Selected Value
Material	Aluminum 6061-T6
Nominal stock dimensions	100 x 40 x 40 mm
X-axis allowance per side	5.55 mm
Y-axis allowance per side	3.49 mm
Approximate sacrificial height	20 mm
Initial top-facing allowance	1 mm
Stock volume	160 cm^3
Approximate stock mass	0.432 kg

The proposed 100 x 40 x 40 mm billet is retained for the preliminary manufacturing study. Its dimensions provide adequate nominal allowances for machining the external geometry and accommodate the sacrificial clamping extension during Setup G54

Final stock dimensions must be verified against supplier tolerances, actual facing requirements and the detailed workholding arrangement.

7.5 STOCK TOLERANCES AND VERIFICATION

The nominal dimension of the Aluminum 6061-T6 stock may vary due to manufacturing and cutting tolerances, which depend on the supplier and stock preparation method. They must be verified before machining.

As established in the previous section, **the minimum theoretical stock height is 35.05 mm**. However, a nominal 35mm billet does not guarantee sufficient material for the finished component, sacrificial extension and initial facing operation. To provide additional allowance, a **nominal height of 40 mm** will be selected for the theoretical manufacturing process. Assuming 1 mm of material removal during initial facing, the available height beneath the finished component becomes **19.95 mm**.

This exceeds the required 15 mm sacrificial extension, providing 4.95 mm for workholding preparation, and it can be removed during subsequent machining operations.

Before machining, the billet dimensions will be verified using a digital caliper, while surface condition, flatness, and suitability for clamping will be inspected. The material grade and temper will be confirmed through supplier documentation. The actual stock height must remain sufficient to accommodate the planed operations, **with 35.05 representing the theoretical minimum before supplier and machining uncertainties are considered.**

The final stock selection is therefore 100 x 40 x 40, aluminum 6061-T6, subject to verification of the supplier's dimensional tolerances.

8. CUTTING TOOLS AND MACHINING PARAMETERS

8.1 MACHINING OPERATIONS AND TOOL REQUIREMENTS

The manufacturing of Bracket 478019 requires a combination of milling, drilling and precision hole-finishing operations. The cutting tools must be suitable for Aluminum 6061-T6, compatible with Haas VF-2SS and capable of achieving the specified tolerances and surface finishes.

The machining operations for each setup are organized as:

Setup	Machining operation	Proposed tool type
G54	Initial stock facing	Carbide face mill or end mill
G54	Rough milling until the first planar face	Ø10 mm, three-flute carbide end mill
G54	Rough milling around the edges	Ø10 mm, three-flute carbide end mill
G54	Finishing of external faces and walls, except internal wall	Carbide finishing end mill
G54	Finishing internal wall	Carbide finishing end mill
G54	Finishing of the internal R0.120 in transition	Small ball-nose end mill with suitable finishing toolpath
G54	Preparation Drilling of center hole	Carbide twist drills
G54	Preparation Drilling of Outer Holes	Carbide twist drills
G54	Precision finishing of the center hole	Adjustable fine-boring system
G54	Precision finishing of the outer holes	Adjustable fine-boring system
G55	Removal of the sacrificial stock extension	Carbide end mill or face mill
G55	Finishing of the underside	Carbide finishing end mill
G56	Drilling of the two flange holes	No. 25 drill
As accessible	Edge breaking and deburring	Suitable deburring tool

Rough milling will use the previously selected Helical Solutions MH35AL-015-31000 solid-carbide end mill (tool No. 59702). The remaining cutting tools will be selected according to their required operations and geometry.

8.2 CUTTING TOOL SELECTION

The cutting tools were selected according to the machining operations in Section 8.1. Considering the properties of aluminum 6061-T6, preference was given to solid-carbide tools designed for aluminum machining, particularly for operations requiring good dimensional accuracy and surface finish. These tools were selected taking into account the component geometry and the capabilities of the Haas VF-2SS machining center.

Primary roughing and Facing Tool

The previously selected Helical Solutions MH35AL-015-31000 is retained for rough milling. This tool will perform the initial roughing of the L-shaped geometry during setup G54. It may also be used for initial facing and selected material-removal operations during Setup G55, reducing the number of separate cutters required.

Finishing Tools

For the finishing of external surface and upright walls, a smaller Ø6 mm solid-carbide end mill is proposed. Helical Solutions tool No. 59794, with three cutting edges and an 18 mm cutting length is selected as a suitable candidate, with the smaller diameter providing greater accessibility around the bracket geometry and allowing separate finishing passes to achieve the specified dimensions and surface quality.

The internal R0.120 in (3.048mm) internal fillet will be machined using a **small ball-nose end mill and a suitable finishing toolpath**. A nominal Ø6 mm ball-nose cutter is proposed, subject to verification of the transition geometry and tool accessibility in Siemens NX CAM.

Precision Hole-Machining Tools

The three mounting holes require strict dimensional control, with a total permissible diameter variation of 0.0254 mm. Therefore, the holes will be drilled undersize and subsequently finished by precision boring.

The proposed preliminary drills are:

Hole	Preliminary Drill	Target Finished Diameter
Two outer mounting holes	Ø9.0 mm	Ø9.5377 mm
Central mounting hole	Ø7.5 mm	Ø7.9121 mm

Preliminary drilling leaves approximately 0.269 and 0.206 mm of radial material for the subsequent boring operation.

A Seco Axiabore A760 02 fine-boring system was chosen for the precision-boring operation for its adjustable range of Ø2-20 mm that accommodates both hole diameters.

The boring head must be equipped with an appropriate small-diameter boring bar and a compatible holder for the selected Haas spindle configuration. The finished hole diameters will be adjusted and verified through dimensional inspection.

The machining sequence must also provide sufficient clearance beneath the base during boring. In Setup G54, the preparatory holes can extend into the sacrificial material, provided that the bore depth, tool clearance and workholding arrangement are verified.

Secondary Drilling and Deburring Tools

The two holes in the upright flange will be machined during Setup G56 using a No. 25 drill, corresponding to a nominal diameter of approximately 3.80 mm. A short, rigid drill suitable for aluminum is preferred to reduce deflection and improve hole positioning.

Finally, an appropriate edge-breaking or deburring tool will be used to remove sharp edges and burrs. The edge treatment must comply with the drawing requirements, **meaning they must be reduced to a radius of 0.01-0.02 in or 0.254-0.508 mm**

Preliminary Cutting Tool List

Tool	Operation	Selected tool or specification
T01	Facing and rough milling	Helical Solution #59702, Ø10 mm, 3- flute carbide (simulated 25mm length provisional)

T02	Wall and surface finishing	Helical Solution #59794, Ø6 mm, 3- flute carbide (simulated 25mm length provisional)
T03	Internal-radius finishing	Ø6 mm carbide ball-nose end mill, provisional
T04	Central-hole preparation	Ø7.5 mm carbide drill
T05	Outer-hole preparation	Ø9.0 mm carbide drill
T06	Precision hole finishing	Seco Axiabore A760 02 fine-boring system
T07	Flange-hole drilling	No. 25 drill, approximately Ø3.8 mm
T08	Edge breaking and deburring	Suitable deburring or edge-finishing tool

The tools listed above constitute the preliminary machining-tool inventory. The final selection of toolholders, boring bars, and remaining manufacturer-specific cutters will be completed before CAM programming. Thei parameters will be established in Section 8.3

Precision Boring Assembly

For the precision finishing of the three mounting holes, the seco Axiabore A760 02 fine-boring system was selected, with an adjustable boring range of Ø2-20 mm. To ensure compatibility with Haas VF-2SS, a Seco EM25024011835 CAT40-to-Graflex G3 adapter is proposed.

Component	Designation	Application
Machine adapter	Seco EM25024011835	CAT40-to-Graflex G3 connection
Fine-boring head	Seco A76002	Adjustable precision boring
Boring bar	Seco A762202	Central hole, Ø7.9121
Boring bar	Seco A763202	Two outer hole, Ø9.5377
Cutting insert	Seco WBGT03102, 03G3	Cutting insert for both bars

The selected carbide boring bars provide sufficient nominal reach for machining through the 6.35 mm base while maintaining clearance from the adjacent upright flange. Their compatibility and accessibility will be verified using the complete tool assembly in Siemens NX CAM

The three holes will initially be drilled undersize, then precision boring will subsequently bring the holes to their target diameters. The same boring head may be used with both provided that it is reset and measured after each tool change.

8.3 CUTTING PARAMETERS

The cutting parameters were established according to the selected tooling, the properties of aluminum 6061-T6, and the capabilities of Haas VF-2SS. The parameters were determined using HSMAdvisor results, manufacturer cutting-data recommendations, and preliminary assumptions for operations where exact tool-specific data remain unavailable.

The main parameters considered are:

- Spindle speed (n)
- Cutting speed (Vc)

- Feed per tooth (fz)
- Feed per revolution (fn)
- Feed rate (Vf)
- Axial depth of cut (ap)
- And radial width of cut (ae)

The formulas used are:

$$n = \frac{(1000 \times Vc)}{\pi \times D} - milling$$

$$Vf = n \times z \times fz - milling$$

$$Vf = n \times fn - drilling\ and\ boring$$

Where D is the tool or machined diameter in mm, z represents the number of cutting edges.

Rough Milling Parameters

The cutting parameters established using HSMAdvisor for Helical Solutions MH35AL-15-31000 carbide end mill:

Parameter	Selected value
Tool diameter	10
Number of flutes	3
Spindle speed	12,000 rpm
Cutting speed	378.8 m/min
Feed per tooth	0.1135 mm/tooth
Feed rate	4,087 mm/min
Axial depth of cut	3.0 mm
Radial width of cut	2.0 mm
Material removal rate	24.5 cm^3/min

The manufacturer's cutting-data chat recommends approx. 0.0965 mm/tooth for rough milling with a comparable 3/8 inch cutter at 30-40% radial engagement.

The selected 2 mm radial engagement represents only 20% of the 10 mm cutter diameter. Considered radial chip thinning, the calculated maximum chip thickness is approximately 0.091 mm, comparable to the chip thickness associated with the manufacturer's recommended roughing conditions.

Therefore, the selected feed rate is considered suitable for preliminary CAM programming, provided that the radial engagement is controlled throughout the toolpath.

Finishing parameters

The Helical Solution #59794, a Ø6 mm three-flute carbide end mill is selected for finishing external surfaces, upright walls, and selected planar features.

To reduce cutting forces and improve dimensional control, smaller radial engagements and feed rates are proposed.

Parameter	Selected value
Spindle speed	10,000 rpm
Cutting speed	188.5 m/min
Feed per tooth	0.05 mm/tooth
Feed rate	1,500 mm/min
Axial depth of cut	3.0 mm
Radial width of cut	0.25 mm

For the internal R3.048 mm transition, a provisional Ø6 mm, two-flute ball-nose end mill is considered. An initial spindle speed of 9,000 rpm, feed rate of 360 mm/min, and 0.2 mm finishing stepover are proposed. The final toolpath and cutter geometry will be verified in Siemens NX CAM.

Drilling Parameters

The three mounting holes will first be drilled undersize before precision finishing. The two smaller flange holes will be drilled during setup G56.

The following preliminary drilling parameters are proposed:

Operation	Drill diameter	RPM	Feed per revolution	Feed rate
Central-hole preparation	7.5 mm	6,000	0.12 mm/rev	720 mm/min
Outer-hole preparation	9.0 mm	5,500	0.14 mm/rev	770 mm/min
Flange-hole drilling	No. 25 (3.797 mm)	4,000	0.05 mm/rev	200 mm/min

Carbide drill suitable for aluminum are assumed for the two larger preparatory holes, while a suitable high-speed-steel drill is considered for the smaller flange holes.

These values are conservative theoretical starting conditions. Feed reduction near drill breakthrough and adequate workpiece support will be considered to limit exit burrs and deformation of the thin features.

Precision Boring Parameters

The three mounting holes will be finished using Seco Axiabore A76002 fine-boring system selected in section 8.2. Because the boring tool uses a single active cutting edge, the feed rate is calculated from spindle speed and feed per revolution. A preliminary cutting speed of approximately 150 m/min and a feed of 0.04 mm/revolution are proposed for the finishing operations.

Parameter	Central hole	Outer holes
Target diameter	7.9121 mm	9.5377 mm
Boring bar	Seco A762202	Seco A763202

Spindle speed	6,000 rpm	5,000 rpm
Cutting speed	149.1 m/min	149.8 m/min
Feed per revolution	0.04 mm/rev	0.04 mm/rev
Feed rate	240 mm/min	200 mm/min

The proposed preparation and finishing sequence is:

Hole	Initial drilling	Final boring
Central	7.50 mm	7.9121 mm
Outer	9.00 mm	9.5377 mm

The remaining radial cutting allowances for fine boring are **0.2061 mm for the central hole and 0.2689 mm for the outer holes.**

These allowances are adopted as preliminary finishing depths. The proposed conditions aim to reduce cutting forces and tool deflection while providing control over the finished diameters.

The designated Ra $\leq$ 1.6 µm surface-finish will be verified through inspection.

Facing and Sacrificial Stock Removal

Initial top facing will be performed using the Ø10 mm end mill, with a preliminary spindle speed of 10,000 rpm, feed rate of 1,800 mm/min and 0.5 mm axial depth per pass.

During Setup G55, the sacrificial stock extension will be removed using successive roughing passes with controlled engagement. A final finishing allowance of approximately 0.3 mm will be retained on the underside before the finishing operation.

The finished base must remain adequately supported during these operations to prevent distortion.

Cutting Conditions and Verification

A suitable water-miscible coolant is proposed for aluminum milling, drilling, and boring to assist chip evacuation and reduce built-up edge formation.

All cutting parameters remain preliminary until verified through Siemens NX CAM simulation and manufacturer tooling data. The final NC program will incorporate appropriate approach movements, machining allowances, and reduced feeds where required by the geometry or workholding conditions.

8.4 TOOLING AND PROCESS VERIFICATION

The selected cutting tools and machining parameters must be evaluated to ensure compatibility with the Haas VF-2SS machining center and the geometrical requirements of Bracket 478019. Particular attention is given to machine operating limits, tool accessibility, dimensional accuracy, and workpiece stability.

Machine and Cutting-Parameter Verification

The proposed roughing operation uses 12,000 rpm and a feed rate of 4,087 mm/min, both with the specified machine limits specified in Section 5. HSMAdvisor predicts a cutting power requirement of 0.28 kW, indicating

a relatively low nominal spindle load. However, the reported cutting torque of 0.77 Nm corresponds to approximately 0.97 kW at 12,000 rpm. Since the calculated and reported power values differ, the predicted loads remain preliminary and require further verification before physical machining.

The Geometry and Accessibility

The selected Ø10 mm roughing end mill has a 15 mm cutting length (25 mm simulated), sufficient for machining the 12.70 mm height from the base and the upright flange using multiple passes. The Ø6 mm finishing end mill provides additional accessibility around the bracket geometry. The internal R3.048 mm (or 0.12 in) transition will be finished using a suitable ball-nose toolpath.

Particular attention is required for the Seco Axiabore A76002 fine-boring assembly. Both precision hole diameters fall within its Ø2-20 mm range, but the complete assembly must be checked for clearance from the upright flange, including the boring head, bars, and toolholder.

The preliminary drilled diameters of 7.50 mm and 9.00 mm leave approximately 0.2061 mm and 0.2689 mm of radial material, respectively, for precision boring. These allowances must be verified against the selected boring-tool capabilities before physical manufacturing.

Workholding and Process Verification

The cutting-force analysis in Section 7.3 established preliminary design loads of 350 N horizontally and 100 N axially. The sacrificial stock extension and proposed clamping arrangements were dimensioned using these theoretical loads.

During rough milling, three programmed radial engagement must remain closed to the selected 2 mm value. Increased engagement at corners or during tool entry could produce higher cutting forces than predicted.

For Setup G55, the thin base must be adequately supported while the sacrificial material is removed. During Setup G56, the upright flange must be securely supported to minimize deformation and burr formation during drilling.

CAM Simulation and Dimensional Verification

The machining operations will be simulated in Siemens NX CAM to evaluate tool accessibility, material removal, and potential collisions between the cutting tools, holders, workpiece and fixtures.

The precision mounting holes will be inspected against their specified diameter ranges

- Outer holes: Ø0.375-0.376 in
- Central hole: Ø0.311-0.312 in

The designated surface finish requirements of Ra $\leq$ 3.2 µm for general machined surfaces and Ra $\leq$ 1.6 µm for precision holes and contact surfaces must also be verified. Inspection will be considered before anodizing, and any subsequent dimensional changes will be assessed separately.

The selected tooling and machining parameters are therefore considered suitable for preliminary manufacturing planning. Final acceptance will depend on successful CAM simulation, verification of the complete tool assemblies, and confirmation of dimensional requirements.

9. DETAILED SIEMENS NX CAM PROGRAMMING

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9.1 CAM ENVIRONMENT AND MACHINE CONFIGURATION

The manufacturing process was developed using Siemens NX 1942.2801, within the Manufacturing application. A general three-axis milling environment was selected for programming the machining operations. The reference CNC machine, Haas VF-2SS was not available in the machine library, therefore a generic machine was selected and modified to represent the principal motion characteristics of VF-2SS.

The machine's linear-axis travel limits and rapid-traverse velocities were configured in the NX machine-tool settings. The resulting axis movements were subsequently tested to confirm the intended directions and travel ranges.

Parameter	Selected configuration
CAM Software	Siemens NX
Machining environment	Three-axis milling
Reference CNC Machine	Haas VF-2SS
NX machine model	Modified generic three-axis milling machine
X-axis travel	762 mm
Y-axis travel	406 mm
Z-axis travel	508 mm
Maximum rapid traverse	35,600 mm/min
Reference maximum spindle speed	12,000 rpm
Reference maximum spindle power	22,4 kW
Reference maximum cutting feed	21,200 mm/min
Reference CNC controller	Haas Next Generation Control (NGC)

The spindle rotary axis was retained in its original configuration, while the required spindle speeds and cutting feeds will be defined individually for each machining operation. The generic machine provides a preliminary representation of the selected machining center, but does not reproduce all of its physical characteristics or controller functions. A compatible Haas postprocessor will therefore be required before the final NC program generation.

9.2 WORKPIECE AND STOCK DEFINITION

A manufacturing assembly was created in Siemens NX to represent the finished component and initial raw material. The assembly contains separate solid models of Bracket 478019 and the rectangular aluminum stock. There are 3 versions of stock and vise, one for each setup. The same model of Bracket is used in all 3 setups and the origin is kept at the center of the part as a reference point in setting up the environment.

The corresponding dimensions are presented below.

Parameter	Dimension
Finished bracket	88.90 x 33.02 x 19.05 mm
Raw stock	100 x 40 x 40 mm
Stock material	Aluminum 6061-T6
Length allowance	5.55 mm per side
Width allowance	3.49 mm per side
Initial top-facing allowance	1.00 mm
Sacrificial material beneath the finished base	19.95 mm

The bracket was centered withing the stock in the two horizontal directions, while its vertical position was established to preserve the sacrificial material required for the clamping arrangement.

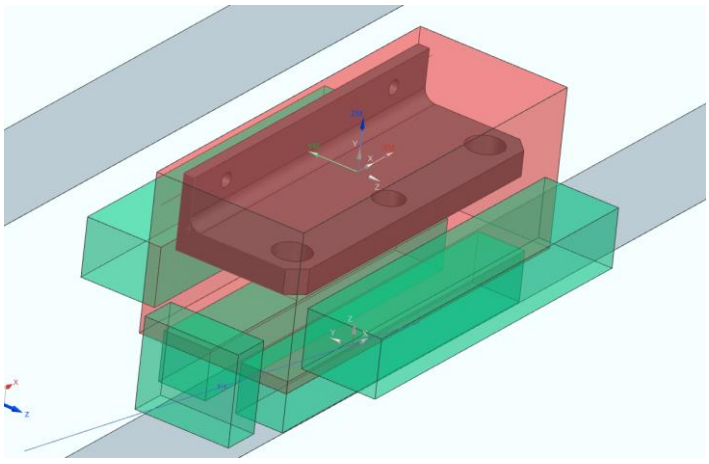
Within the NX Manufacturing environment, the finished bracket was assigned as Part Geometry, while the rectangular billet was assigned as Blank Geometry.

This configuration allows NX CAM to distinguish between the target component and the initial material volume, providing the basis for subsequent toolpath generation and material-removal simulation.

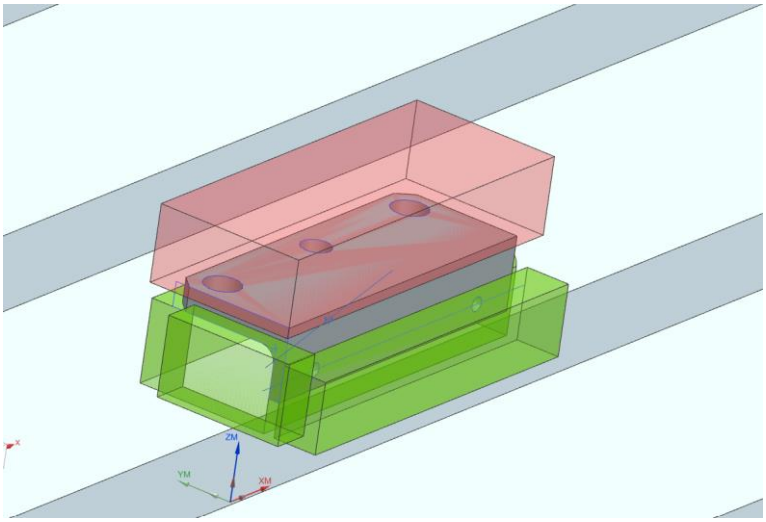
9.3 MACHINING COORDINATE SYSTEMS AND SETUPS

The manufacturing process was organized into three setups, each requiring a separate machining coordinate system (MCS) and workholding configuration.

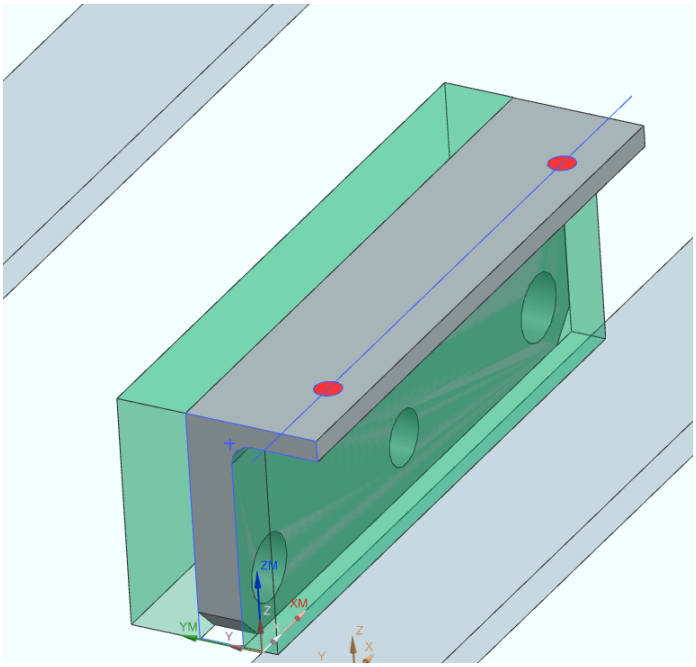
In G54, the initial billet was secured through the sacrificial material beneath the finished bracket, allowing the milling of the L-shape geometry and the drilling of the three larger mounting holes using a simple vise model.



For G55, the component was inverted 180° to expose and remove the sacrificial material, then finish the surface. A supporting fixture clamping the thin wall and providing a base underneath the L-shape was used to maintain stability during machining.



Finally, G56 involved repositioning the bracket to align the two smaller flange holes with the spindle axis for drilling, providing support while maintaining a clear space underneath the flange for allowing the drill to pass.



Separate stock and fixture assemblies were created in Siemens NX for each setup, while the same finished CAD model served as the geometric reference.

9.4 CUTTING TOOLS AND CAM PROGRAMMING STRATEGY

The cutting tools selected in chapter 8 were defined in the Siemens NX tool library and assigned to the corresponding machining operations, using the parameters available in NX.

The Ø10 mm and Ø6 mm carbide end mills were modeled with 25 mm cutting lengths to ensure accessibility during external machining and sacrificial-stock removal. These modified geometries represent the simulation tools and require verification against commercially available cutters.

Machining operations were organized according to their respective program groups, tools, workpiece geometry and machining methods.

The general strategy consisted of rough milling with finishing allowances, subsequent surface finishing, drilling of undersized mounting holes, and precision boring to the required diameters.

The remaining stock was considered during successive machining operations to prevent unintended material removal and preserve the workholding regions.

9.5 SETUP G54: PRIMARY MACHINING OPERATIONS

The first setup included the principal roughing, finishing, drilling and boring operations.

Initial facing was performed using the Ø10 mm carbide end mill to establish a planar reference surface, then the L-shaped geometry was produced using two CAVITY_MILL operations: the first removed material above the horizontal base, while the second machined the external perimeter.

The roughing parameters were established at 12,000 rpm, 4,087 mm/min feed rate, 3mm maximum axial depth and 2mm nominal stepover. Finishing allowances of 0.5 mm on walls and 0.3 mm on floors were retained were applicable.

The cutting range of the internal roughing operation was restricted to the upper surface of the horizontal base, preventing unnecessary machining of the sacrificial material beneath it.

The external and internal walls were subsequently finished using the Ø6 mm end mill and FLOOR_WALL operations. A separate finishing operation was required for the internal upright wall because it was not completely machined by the initial finishing toolpath.

The internal R3.048 mm fillet was finished using a Ø6 mm ball-nose end mill and the CONTOUR_AREA operation.

The finishing parameters were:

Operation	RPM	Feed (mm/min)
Wall and surface finishing	10,000	1,500
Internal-radius finishing	9,000	360

The three mounting holes were prepared using Ø7.5 mm and Ø9.0 mm drills, then programmed for precision boring using the Seco Axiabore system configured in previous sections.

Hole	Preliminary diameter	Final diameter	Boring RPM	Feed (mm/min)
Central	7.50 mm	7.9121 mm	6,000	240
Outer(2)	9.00 mm	9.5377 mm	5,000	200

The boring operations were assigned to BORE_METHOD. The G54 Simulation demonstrated the progressive formation of the L-shaped component while preserving the sacrificial stock required for the following setup.

9.6 SETUP G55: REVERSE-SIDE MACHINING

After completing the primary machining operations, the bracket was inverted and secured in the second fixture arrangement. Two operations were developed: roughing the sacrificial stock and finishing the remaining component face.

The Ø10 mm end mill was used to remove the approximately 19.95 mm sacrificial extension through successive cutting passes, retaining approximately 0.30 mm for final finishing.

The Ø6 mm end mill was subsequently used to remove the remaining allowance and produce the finished underside surface. The finishing operation targeted a horizontal-base thickness of 6.35 mm, while the supporting fixture maintained workpiece stability during removal of the remaining material. The Siemens NX simulation showed the gradual removal of the sacrificial extension and the formation of the final underside geometry.

9.7 SETUP G56: PERPENDICULAR HOLE DRILLING

Setup G56 was developed to produce the two No. 25 through-holes located on the upright flange as specified in the original drawing. The bracket was repositioned so that the hole axes were parallel to the spindle Z-axis and secured using a supporting fixture.

The programmed drilling parameters were:

Parameter	Value
Tool diameter	3.797 mm
Tool type	No. 25 HSS twist drill
Spindle speed	4,000 rpm
Feed per revolution	0.05 mm/rev
Feed rate	200 mm/min
Number of holes	2

The drilling depth was defined to achieve the complete breakthrough of the upright flange while maintaining sufficient clearance from the supporting fixture.

The operation completed the principal CNC machining sequence of Bracket 478019.

9.8 CAM PROGRAMMING VERIFICATION

Following toolpath generation, the machining operations were reviewed using the Siemens NX toolpath visualization and material-removal simulation functions. The G54, G55 and G56 setups were simulated separately to account for their different coordinate systems, stock conditions and fixture arrangements.

During CAM development, several toolpath issues were identified and corrected, including the tool length which was initially set at 15 mm and did not allow for machining without collisions. Other issues included incomplete external material removal and the omission of the internal wall from the initial finishing operation.

These corrections resulted in a revised machining sequence incorporating separate external roughing and internal-wall finishing operations.

The resulting simulations provided visual confirmation of the intended manufacturing sequence and the proper removal of material, confirming the proposed strategy.

Additional IPW and collision verification was performed in Siemens NX, as presented in Section 10. Nevertheless, complete machine-level verification and physical dimensional inspection remain necessary before the manufacturing process can be considered production ready.

10. MACHINING VERIFICATION AND SIMULATION

10.1 SIMULATION METHODOLOGY

The machining operations developed in Chapter 9 were evaluated using the toolpath visualization and material removal simulation function in Siemens NX. The objective was to examine the machining sequence, verify the intended material removal, assess tool accessibility and identify potential problems involving the workpiece and fixtures.

The three machining setups, G54, G55 and G56, were evaluated separately, given that each necessitated a different workpiece and workholding setup.

The simulations were carried out using the programmed cutting tools and machining parameters presented in past sections, **with the In-Process Workpiece (IPW) representing the evolution of the material during successive operations.** Visual verification was performed using the Siemens NX 3D Dynamic simulation, allowing the tool movements and progressive material removal to be examined.

10.2 G54 SIMULATION RESULTS

The first setup involved the most complex machining sequence, including facing, internal and external roughing, surface finishing, internal-radius finishing, drilling and precision boring. During initial simulation, several toolpath problems were identified and corrected.

The original roughing operation was found to not be sufficient for the complete removal of the surplus material, therefore, another roughing operation was performed around the vertical faces.

During wall finishing, the internal upright wall was not completely machined by the initial FLOOR_WALL operation, therefore another operation for finishing the internal wall of the L-shape was performed.

Following these adjustments, the simulation showed the progressive removal of material to reveal the L-shaped. Subsequently, the three mounting holes were initially drilled undersize and subsequently finished using precision borind operations targeting the specified nominal diameters.

The sacrificial stock beneath the bracket was retained, allowing the component to remain supported throughout the first setup.

The resulting geometry appeared consistent with the intended manufacturing strategy.

10.3 G55 AND G56 SIMULATION RESULTS

The second setup focused on removing the sacrificial material retained during G54. The NX simulation illustrated the progressive removal of the material using the same Ø10 mm and Ø6 mm mills used in setup G54.

The resulting virtual workpiece showed the exposed underside of the horizontal base and the three previously machined mounting holes.

This confirmed the intended sequence of removing the temporary stock after repositioning. The final base thickness were evaluated using IPW analysis and must be subjected to further verification in case of physical production.

For G56, the bracket was repositioned with the two No. 25 holes to align with the spindle axis. The drilling operation was programmed using a Ø3.797 mm twist drill with a spindle speed of 4,000 rpm and a feed rate of 200 mm/min.

The setup geometry provides access to both hole locations, the fixture leaving empty space under the holes but supporting the upright flange using a custom setup.

10.4 VERIFICATION RESULTS AND LIMITATIONS

The Siemens NX simulations provided a preliminary assessment of the proposed manufacturing process and supported the development of a feasible three-setup machining strategy. The primary results are summarized in the following table:

Verification Criterion	Current Assessment
G54 machining sequence	Visually simulated
L-shaped geometry formation	Visually confirmed
External roughing	Toolpath corrected and simulated
Internal wall finishing	Additional operation introduced
G55 sacrificial-stock removal	Visually simulated
G56 drilling operation	Visually simulated
Residual stock analysis	IPW analysis performed; no significant unwanted material identified on the inspected surface.
Tool and fixture collision analysis	No collisions observed during visual simulation; complete machine-level verification remains necessary.
Dimensional accuracy	Virtual geometry appears consistent with the CAD model; tolerance conformity is not fully established.
Machining time	7 min 57 s

The simulation process allowed several programming errors to be identified and corrected before physical manufacturing. Nevertheless, the available results are primarily based on visual toolpath and material-removal verification.

Furthermore, the modified generic three-axis machine used in Siemens NX does not fully reproduce the physical and control characteristics of the Haas VF-2SS. Consequently, the simulated process must be considered a preliminary virtual manufacturing validation rather than evidence of production readiness.

11. DIMENSIONAL INSPECTION AND QUALITY CONTROL

11.1 INSPECTION REQUIREMENTS

Dimensional inspection will be performed on Bracket 478019 to ensure conformity with its engineering drawing. The inspection plan will consider the initial NASA/Essex Corporation tolerances and the additional surface roughness added in this project.

Main inspection characteristics:

Characteristic	Specification	Inspection Method
Overall dimensions	3.50 x 1.30 0.75 in	Caliper / micrometer

Horizontal base thickness	0.250 in	Micrometer
Upright wall thickness	0.120 in	Micrometer
Two larger holes	Ø0.375-0.376 in	Bore gauge / pin gauges
Central hole	Ø0.311 -0.312 in	Bore gauge / pin gauges
Two smaller holes	2 x No.25 drill, nominal Ø0.149 in	Pin gauges
Internal fillet	R0.120 in	Radius gauge / optical comparator
Hole positions	As specified on drawing	CMM / height gauge
General surface roughness	Ra ≤ 3.2 µm (125 µin)	Surface profilometer
Functional surfaces and precision holes	Ra ≤ 1.6 µm (63 µin)	Surface profilometer
Edge treatment	R0.01-0.02 in	Optical inspection / radius gauge

The original drawing specifies the following tolerances, where applicable

- **General dimensional tolerances:**
  - Two-decimal dimensions: ±0.015 in
  - Three-decimal dimension: ±0.005in
- **General angular tolerance:** $\pm 1^{o}$

Explicit dimensional limits take precedence over general tolerances. All measuring equipment must have adequate accuracy and calibration.

11.2 INSPECTION PROCEDURE

Inspection will be performed at three stages of the manufacturing process:

1. Raw Material Inspection

The aluminum billet will be inspected before machining to verify its dimensions, surface condition, and material specification and ensuring sufficient material allowances for machining and workholding.

2. In-Process Inspection

Following each machining setup, available critical dimensions will be inspected before repositioning, including wall thickness, machined reference surfaces, and hole diameters.

The closely toleranced hole will require particular attention to ensure their dimensions remain within the specified 0.001-inch diameter range.

Surface roughness will be verified before anodizing and after finishing operations, using a suitable profilometer.

3. Final Inspection

Following machining and deburring, the bracket will undergo a complete dimensional inspection against the technical drawing, verifying dimensional conformity, hole positioning, surface roughness, edge treatment and the absence of visible machining defects.

11.3 QUALITY CONTROL AND DOCUMENTATION

A simplified First Article Inspection (FAI) report template is proposed, **following the principles of SAE AS9102C:2023. The report will be completed after physical manufacturing and dimensional inspection.**

The report will identify:

- each inspected characteristic
- its nominal dimension
- allowable tolerance
- measured value
- conformity status.

Any deviation from the specified requirements will be recorded and evaluated before component acceptance.

This project is educational, therefore, the inspection plan represents a proposed quality-control procedure. Actual dimensional compliance can only be confirmed after the manufacturing and measurement of the physical component.

12. FINISHING AND POST-PROCESSING

12.1 DEBURRING AND EDGE TREATMENT

Following CNC machining, all burs and sharp edges will be removed in accordance with the original NASA/Essex engineering drawing.

The drawings specifies an edge-break radius of R0.01-0.02 inches (0.254-0.508 mm) This requirement will be achieved through controlled manual deburring or suitable edge-finishing tools. Particular attention will be given to drilled holes, external corners, and the internal fillet. Deburring must not compromise the dimensional tolerances or surface quality of the finished component.

12.2 SURFACE CLEANING

Before surface treatment, the component will be cleaned to remove machining coolant, lubricant, chips, and other contaminants.

Cleaning will be performed using a suitable aluminum-compatible degreasing process. The component will then be inspected to ensure that all machined surfaces are free of contamination and visible defects.

The specified surface roughness requirements of $Ra \leq 3.2$ µm for general surfaces and $Ra \leq 1.6$ µm for designated functional surfaces will be verified before anodizing.

12.3 PROTECTIVE SURFACE TREATMENT

The original engineering drawing specifies a clear anodized finish for the aluminum component.

For the proposed manufacturing process, sulfuric acid anodizing according to MIL-PRF-8625F, Type II, Class 1, is selected as a modern reference specification. This process produces an undyed protective aluminum oxide layer, improving corrosion resistance and surface durability.

Anodizing will be performed by a specialized surface-treatment provider, followed by an appropriate sealing process.

The anodic coating may affect component dimensions, particularly the closely toleranced holes (Ø0.375-0.376 in and Ø0.311-0.312 in). Therefore, coating thickness and potential dimensional changes must be considered before treatment.

The treatment of the precision holes will be determined in consultation with the surface-treatment provider. Depending on the coating requirements, masking or adjustements to the pre-anodizing machining dimensions may be necessary to ensure that the final hole diameters remain within the specified tolerances.

The finished component will undergo dimensional verification after anodizing to confirm conformity with the original engineering drawing.

12.4 FINAL INSPECTION

After anodizing, the component will be visually inspected for coating uniformity, discoloration, surface defects, and damage.

Critical dimensions, particularly precision hole diameters, will be rechecked to verify conformity with the engineering drawing.

The completed component will then be cleaned, protected against handling damage, and prepared for storage or presentation.

13. MANUFACTURING RISK AND SAFETY

13.1 CNC MACHINE AND OPERATOR SAFETY

The proposed physical manufacturing of Bracket 478019 is based on the Haas VF-2SS machining center. Therefore any future production must follow the manufacturer's safety instructions and applicable machine-tool safety requirements, including ISO 16090-1:2022.

The principal safety measures include:

- Verifying machine condition, and setup installation, including tool and workholding security before the start of operations
- Verifying the functionality of the safety interlocks and protective enclosure during operations
- Using appropriate personal protective equipment at every stage of the manufacturing process
- Verifying process parameters, like work offsets, tool lengths and NC programs before executing machine operations
- Using controlled program prove-out procedures to reduce the risk of unexpected machine movements
- Keeping the hands and measuring instruments away from moving machine components

13.2 MANUFACTURING RISK ASSESSMENT

The main manufacturing risks are associated with the rigidity of the workpiece, dimensional accuracy, tool selection, and the multiple machining setups required by the component geometry. The manufacturing process must take into account the possibility that splinters might appear, or that a non-conform workspace setup

could lead to the damage of the machine, tools, of the workpiece or any other entity involved in the manufacturing process.

Potential risk	Possible consequence	Preventive measure
Insufficient clamping force	Workpiece movement or ejection	Verify fixture rigidity and required clamping force
Excessive clamping pressure	Deformation of thin walls	Use supporting fixtures and distribute clamping loads
Excessive cutting forces	Tool deflection, vibration, dimensional errors	Optimize cutting parameters and finishing allowances
Incorrect work offsets	Tool collision or machining errors	Verify G54, G55, and G56 before machining
Inadequate tool clearance	Collision with fixtures or workpiece	Perform collision verification in Siemens NX
Tool wear or runout	Poor surface finish and dimensional inaccuracies	Inspect tools and use suitable finishing strategies
Drilling breakthrough	Burr formation or fixture damage	Provide adequate drill-exit clearance
Anodizing dimensional changes	Precision holes outside tolerances	Account for coating growth and verify final dimensions

The dimension of the bracket imposes a very strict tolerance. The thin walls and the precision holes require a rigid workholding setup that also ensure the clamping forces don't lead to deformation or the breaking of the part.

13.3 RISK MITIGATION AND PROCESS VALIDATION

The proposed manufacturing process will be evaluated through Siemens NX CAM simulation to identify potential tool collisions, unmachined material and machining setup errors.

Before physical machining, cutting forces and workholding requirements will be assessed to confirm that the selected stock and fixtures provide adequate support.

The generated NC program will require verification on the actual machine, and initial machining will be performed using controlled feed rates and appropriate program prove-out procedures. Dimensional inspections will be performed between setups to identify manufacturing deviations before subsequent operations.

The preliminary risk assessment incorporates the selected cutting tools, calculated machining parameters and Siemens NC simulation results. Further validation of the physical workholding arrangements, cutting tools and generated NC programs will be required before manufacturing.

14. MANUFACTURING TIME AND COST ANALYSIS

14.1 MACHINING TIME ESTIMATION

The manufacturing time of Bracket 478019 was estimated using the machining-time calculation function available in Siemens NX, based on the simulation presented in Section 10. The machining operations were divided

into three setups, G54, G55 and G56, each involving different workpiece orientations and manufacturing operations. The estimated machining times obtained are:

Setup	Main operations	Machining time
G54	Facing, rough milling, finishing, drilling and boring	4 min 49s
G55	Sacrificial-stock removal and underside finishing	2 min 51 s
G56	Two No. 25 through-holes	17 s
Total	Three machining setups	7 min 57 s

The total estimated machining time is therefore 477 seconds or

$$\mathrm{T} = 289 + 171 + 17 = 477\mathrm{s}$$

$$\mathrm{T} = 477/3600 = 0.1325\mathrm{h}$$

This value represents the machining time estimated by the Siemens NX CAM software and does not include workpiece preparation, repositioning between setups, dimensional inspection or post-processing.

Furthermore, the modified generic three-axis machine used during simulation does not fully reproduce the behavior of the reference Haas VF-2SS. Consequently, actual manufacturing time may differ from the simulated results.

14.2 MANUFACTURING COST ESTIMATION

The manufacturing cost of Bracket 478019 can be estimated by considering the raw material, CNC operation machine operating time, cutting tools, setup procedures, dimensional inspection and post-processing operations. The objective is to establish a preliminary cost per component using the results obtained from simulation and representative industrial cost assumptions.

Raw Material Cost

The selected aluminum 6061-T6 billet has dimensions of 100 x 40 x 40 mm, resulting in a volume of 160 cm^3.

Using the approximate density of 2.70 g/cm^3, the mass is 0.432kg. At the moment, a rough estimate price per kg for aluminum 6061-T6 is €5. **Therefore, the total material cost can be estimated at: €2.16.**

CNC Machining Cost

The CNC machining cost depends on the machine operating time and the hourly operating rate. Assuming an illustrative operating rate of €60/hour for our Haas VF-2SS (this is not a direct quotation), the corresponding costs for each machining setup are:

Setup	Machining time	Estimated cost
G54	4 min 49s	€4.82
G55	2 min 51 s	€2.85
G56	17 s	€0.28
Total	**7 min 57 s**	**€7.95**

An actual hourly operating rate would account for machine depreciation, maintenance, electrical energy consumption, operator costs where applicable and manufacturing overhead, thus it can vary between different countries, workshops, enterprises etc.

Setup and Tooling Costs

The manufacturing process involves three separate setups, each requiring workpiece positioning, fixture preparation and machining-coordinate verification. These operations introduce additional costs that must be considered independently from the estimated cutting time.

For a production batch, the setup allocated to each component is calculated as:

Setup Cost per Part = (Total Setup Time x Setup Hourly rate) / Number of Parts

For example, assuming, a total setup time: 1 hour, Setup hourly rate: €40/hour and Production quantity: 10 brackets, we get a **Setup Cost per Part = (1 x 40)/10 = €4.00**

Tooling costs must also be considered, particularly for carbide end mills, drills and precision-boring equipment selected in Section 8. Since cutting tools are generally reusable, their cost can be distributed over the number of components manufactured before replacement.

Tool Cost per Part = Tool Price / Number of Parts Machined Before Replacement.

This calculation is performed for each cutting tool, and the individual costs are added together. A more precise estimate would require actual tool prices, expected tool life and the number of cutting edges available for use.

Inspection and Post-Processing Costs

Additional manufacturing expenses include dimensional inspection, surface finishing, deburring, cleaning and clear anodizing. The inspection cost depends on the time required to verify the bracket's dimensions, hole diameters and surface quality, together with the equipment used.

The post-processing cost includes the removal of burrs, treatment of sharp edges and application of the protective clear anodized finish specified in the original engineering drawing. These costs would normally be determined using estimated processing times or quotations from specialized manufacturing providers.

Total Manufacturing Cost

The theoretical manufacturing cost per component is determined by adding the individual cost categories:

Total Cost per Part = Material Cost + Machining Cost + Setup Cost per Part + Tooling Cost per Part + Inspection Cost + Post-processing Cost

Using the illustrative values calculated previously:

Cost category	Estimated cost
Raw material	**€2.16**
CNC machining	**€7.95**
Setup cost per part (10-part batch)	**€4.00**
Cutting-tool consumption	**To be determined**
Dimensional inspection	**To be determined**

Post-processing and anodizing	**To be determined**
Stock preparation	**To be determined**
Calculated subtotal	**€14.11**

Therefore, under the selected assumptions, the preliminary manufacturing cost is **€14.11 per bracket, excluding tooling, inspection, stock preparation and post-processing expenses.** The final manufacturing cost will depend on production quantity, supplier prices, actual machine operating rates and the additional operations required.

The analysis demonstrates that manufacturing costs are influenced not only by machining time, but also by setup preparation, tooling and finishing requirements. **The requirement of using multiple fine tools for reaching the required dimensional values and the need of three different setups push the cost significantly, as the duration of changing the setup is much more significant than the cost of operating the machine**.

15. MANUFACTURING RESULTS AND CONCLUSIONS

15.1 MANUFACTURING RESULTS

The Manufacturing Process Plan for NASA Bracket 478019 established a proposed CNC manufacturing strategy based on the original engineering documentation.

The component was reconstructed in Siemens NX, and its geometry, material specifications, dimensional tolerance and manufacturing requirements were analyzed.

A three-setup manufacturing process was developed using a modified generic three-axis CNC milling machine , with the Haas VF-2SS serving as the reference equipment.

The process incorporated initial facing, internal and external rough milling, surface finishing, internal radius finishing, drilling and precision boring, followed by sacrificial-stock removal and perpendicular hole machining.

The machining operations were simulated using Siemens NX, allowing the manufacturing strategy to be evaluated and several programming issues to be identified and corrected.

The In-Process Workpiece analysis was also used to examine the resulting material-removal geometry and identify possible deviations from the finished CAD model.

The manufacturing process produced the intended component geometry in the virtual environment, although the available simulation results do not independently establish complete dimensional or surface-finish conformity.

Parameter	Result
Component	NASA Bracket 478019
Material	Aluminum 6061-T6
Finished dimensions	88.90 x 33.02 x 19.05
Initial stock	**100 x 40 x 40 mm**
Reference CNC machine	**Haas VF-2SS**
CAM software	**Siemens NX**
Manufacturing setups	**3**

Simulated machining time	**7 min 57 s**
Illustrative machining cost	**€7.95**
Preliminary cost subtotal	**€14.11 per part (10-part batch assumptions)**

The machining-time and cost calculations provided an additional assessment of the proposed process, demonstrating the relationship between cutting time, setup requirements and total manufacturing expenses.

15.2 CONCLUSIONS

The project demonstrated the application of CAD/CAM and manufacturing engineering principles to the reconstruction and proposed production of an aerospace-related mechanical component. The integration of engineering drawing analysis, material selection, cutting-tool selection, machining-parameter calculations, workholding planning and CNC simulation provided a structured approach to manufacturing process development.

Several manufacturing challenges were identified during the project, including the machining of thin walls, preservation of sacrificial clamping material, internal-radius finishing and the preparation of precision mounting holes.

The simulation process demonstrated the importance of verifying machining operations individually and modifying the manufacturing strategy when toolpath limitations were encountered.

The estimated machining time of approximately 7.95 minutes demonstrated that the component can be programmed for a relatively short CNC machining cycle. However, the manufacturing cost analysis also highlighted the importance of setup preparation, tolling and post-processing expenses, particularly for small production quantities.

Although the virtual manufacturing process established a technically plausible machining sequence, additional verification remains necessary before physical production.

Future work should include complete collision verification, validation of machining parameters using the final cutting tools, Haas-compatible NC program verification and physical manufacturing followed by dimensional and surface-quality inspection.

Overall, the project achieved its principal objective of developing and documenting a technically justified CNC manufacturing process for NASA Bracket 478019 using Siemens NX, providing a basis for further manufacturing optimization and physical validation.